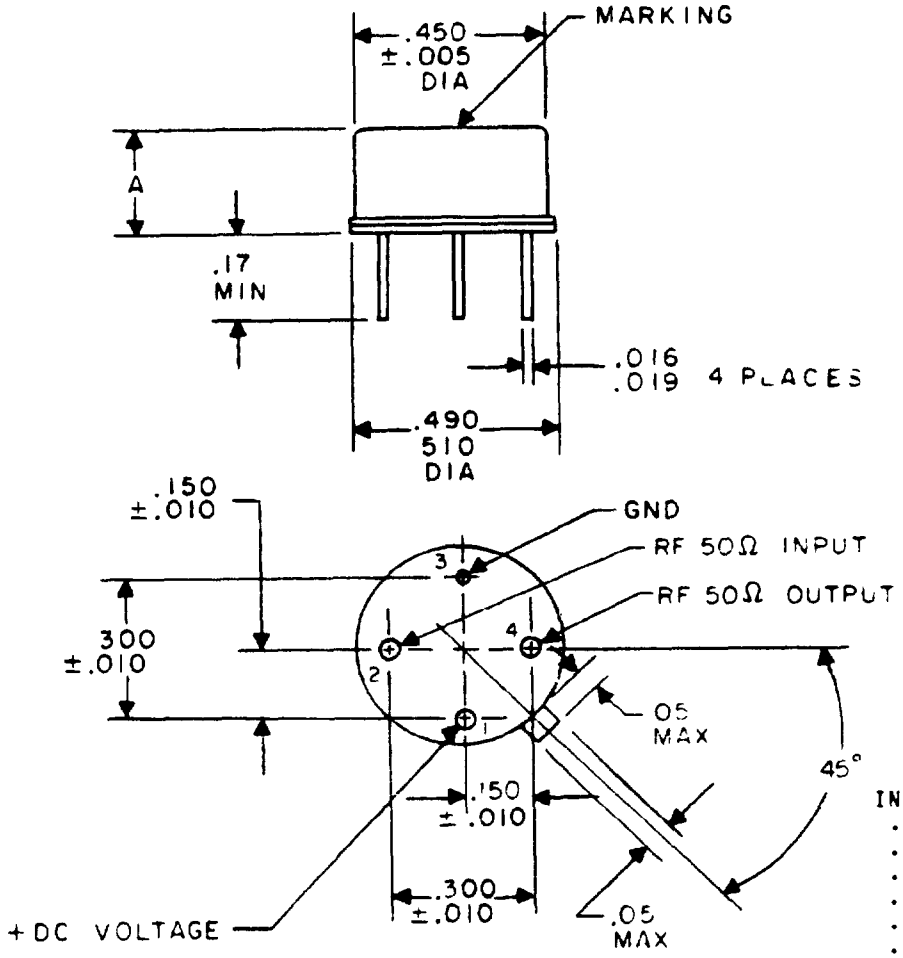
MILITARY SPECIFICATION SHEET

AMPLIFIERS, RADIOFREQUENCY AND MICROWAVE, SOLID-STATE, LOW NOISE, WIDEBAND, TO CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring amplifiers described herein shall consist of this specification sheet and the latest issue of MIL-A-28875.



INCHES	MM
.005	.13
.010	.25
.016	.41
.019	.48
.05	1.3
.150	3.81
.17	4.3
.18	4.6
.22	5.6
.27	6.9
.300	7.62
.450	11.43
.490	12.45
.510	12.95

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

FSC 5085

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TABLE I. Electrical characteristics (25°C unless otherwise specified).

Dash number	Frequency range (MHz)	DC input power: Voltage	DC input power: Current Max (mA)	Gain (dB) Min	Gain flatness (dB) Max	RF output power (dBm) Min	VSWR Max: IN	VSWR Max: OUT	Noise figure (dB) Max	Intercept point 3rd order (dBm) Min	Harmonic output (dBc)	Gain vs temperature (dB)	RF input power (dBm) Max	Isolation (dB) Min	Phase linearity (deg) Max
S1	5-250	+15 ±1%	40	31	±0.2	+9	1.3:1	1.3:1	4						
01 S2				29 from -54°/+85°C	1.0 from -54°C/+85°C	+7 from 54°C/+85°C	2.0:1 from +85°C	2.0:1 -54°C	5.5 from -54°C/85°C	+21	---	±2	+17	---	---
N															
S1	200-400	+15 V dc ±1 V dc	20	12.2	±0.5	+6	1·17:1	1.17.1	2.5	+19	---	---	+13	25	±.5
02 S2															
N															
S1	5-500	+15 ±1%	25	20	±1.0	+7	2 0·1	2.0:1	4.5	+20	---	---	---	---	---
03 S2															
N															
S1	5-500	+15 ±1%	10	15	±1.0	-2	2.0:1	2.0 1	2.5						
04 S2						-3 from -54°/+85°C			3.0 from -54°/85°C	+8	---	---	+13	---	---
N															
S1	5-500	+15 ±1%	10	14	±1.0	-2	2.0.1	2.0·1	4						
05 S2				13.5 from -54/85°C		-3 from -54°/+85°C			4.5 from -54°/+85°C	+11	---	±0.5	+13	---	---
N															
S1	5-500	+15 ±1%	23	20	±1.0	+7	2.0:1	2.0.1	4.5						
06 S2				19 from -54/+85°C					5.0 from -54°/+85°C	+20	---	±1.0	+13	---	---
N															
S1	5-500	+15 ±1%	23	14	±1.0	+7	2.0:1	2.0·1	5.5						
07 S2				13.5 from -54/+85°C					6.0 from -54°/+85°C	+21	---	±0.5	+13	---	---
N															
S1	2-1000	+15 ±1%	10	14	±0.7	-5	2.0·1	2.2·1	3.5						
08 S2				13.5 from -54/+85°C	±1.0 from -54/+85°C	-6 from -54/+85°C			4 from -54°/+85°C	+10	---	---	+13	---	---
N															
S1	5-1000	+15 ±1%	25	14	±1.0	+7	2.0:1	2.0 1	6.5						
09 S2				13 from -54/+85°C	±1.2 from -54/+85°C	+6.5 from -54/+85°C			7 from -54°/+85°C	+21	---	---	+17	---	---
N															

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TABLE I. Electrical characteristics (25°C unless otherwise specified) - Continued.

Dash number	Frequency range (MHz)	DC input power Voltage	DC input power Current Max (mA)	Gain (dB) Min	Gain flatness (dB) Max	RF output power (dBm) Min	VSWR Max IN	VSWR Max OUT	Noise figure (dB) Max	Intercept point 3rd order (dBm) Min	Harmonic output (dbc)	Gain vs temperature (dB)	RF input power (dBm) Max	Isolation (dB) Min	Phase linearity (deg.) Max
S1	10-1200	+15 ±1%	40	24	±0.8	+7	2.0 1	2.0:1	4.3						
10 S2				23 from -54/+85°C	1.0 from -54/+85°C	+6.5 from -54/+85°C			4.8 from -54°/+85°C	+20	---	---	+17	---	---
N															
S1	10-2000	+15 ±1%	12	10.5	±1.0	-4.5	2.2 1	2.2·1	4.5						
11 S2				from -54/+100°C	from -54/+100°C	from -54/+100°C			from -54°/+100°C	+10	---	---	+27	---	---
N															
S1	100-2000	+15 ±1%	40	15	±0.7	+6	2.0.1	2.0:1	6.5						
12 S2				14 from -54/+85°C	+1.0 from -54/+85°C	+5 from -54/+85°C			7.0 from -54°/+85°C	18	---	---	+17	---	---
N															

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TABLE II. Dimension, physical and environmental characteristics.

Dash number	Dimension A in inches (mm) Max	Weight Max (Grams)	Temperature range		Burn-in temperature (case temp)	Seal	Salt
			Operating case temp	Nonoperating			
01 S1 S2 N	.22	2.5	-54/+85°C	-65/+100°C	+100°C	Hermetic	Applicable
02 S1 S2 N	.27	2.5	-55/+100°C	-65/+125°C	+100°C	Hermetic	Applicable
03 S1 S2 N	.18	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
04 S1 S2 N	.18	2.5	-54/+85°C	-65/+100°C	+100°C	Hermetic	Applicable
05 S1 S2 N	.18	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
06 S1 S2 N	.18	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
07 S1 S2 N	.18	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
08 S1 S2 N	.18	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
09 S1 S2 N	.18	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
10 S1 S2 N	.22	2.5	-54/+85°C	-65/+100°C	+100°C	Hermetic	Applicable
11 S1 S2 N	.22	2.5	-54/+100°C	-62/+125°C	+125°C	Hermetic	Applicable
12 S1 S2 N	.22	2.5	-54/+100°C	-62/+125°C	+125°C	Hermetic	Applicable

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table II.

Electrical characteristics: See table I.

Physical and environmental characteristics: See table II.

Part number. M28875 /5-(dash number from tables I and II).

NOTE: Amplifiers covered by this specification sheet are substitutable for the following manufacturer's or user's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash number	Manufacturer's code	Manufacturer's part number
01S	49956	954498-1
01S	33875	EL-F0034-002
01S	14482	WJ-A74-1
02S	91417	132845-101
02S	55027	QBH-309
03S	15280	82-052029-02
03S	24539	ASF9054
04S	57958	5054882-3
04S	24539	UTO-511R
05S	57958	5054882-6
05S	24539	UTO-501R
06S	57958	5054882-4
06S	53711	5399796-1
06S	24539	UTO-512R
07S	57958	5054882-5
07S	24539	UTO-502R
08S	24539	UTO-1011R
09S	91417	132851-101
09S	14482	WJ-A15
09S	24539	UTO-1002R
10S	52340	72-00003-001
10S	14482	WJ-A64
11S	06401	7912059-01
11S	14482	WJ-A31-1
12S	14482	WJ-A34

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities.
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities·
Army - AV
Navy - AS, MC
Air Force - 19

Agent
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0968-5)

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